Karlshochschule
International University

Bachelor Thesis

"The impact of the Covid-19 pandemic on the consumer behavior of online shopping in Germany"

International Business (Intercultural Management)

Supervisor: Prof. Dr. Wendelin Küpers

Submitted by

Woo-Sung Hollenhorst

H18201-015

Bensheim, 28.05.2024

Table of Contents

Statutory Declaration 3

1. Abstract 4

2. Introduction 6

    2.1 E-Commerce market of Germany 7

    2.2 Online shopping behavior in Germany 7

    2.3 Changes of businesses during the pandemic 9

3. Research introduction 10

4. Methodology 11

    4.1 Quantitative research design 11

    4.2 Qualitative research design 15

5. Results 18

    5.1 Quantitative results 18

    5.2 Qualitative Results 27

6. Implications 30

7. Conclusion 33

8. Limitations and further research directions 34

9. References 36

10. Appendix 39

Statutory Declaration

I herewith declare that I have authored the present assessment independently making use only of the specified literature. Sentences or parts of sentences quoted literally are marked as quotations; identification of other references with regard to the statement and scope of the work is quoted. The assessment in this form or in any other form has not been submitted to an examination body and has not been published.

1. Abstract

The Covid-19 pandemic had an everlasting effect on humanity and brought major changes of different aspects within society. Consumer behavior was one of the most significantly affected domains.

This research will explore the impact that the pandemic had on online shopping behavior within Germany. The aim of this study is to elucidate the changes of consumer patterns, preferences and motivations of the online shopping sphere before, during and post-pandemic.

Through the combination of surveys, interviews and empirical data of reputable sources, the study employs a mixed-methods approach, combining qualitative and quantitative research methodologies to provide a comprehensive analysis of the pandemic's effects, trends and correlations of the focus group can be made. A survey conducted on a broader sample of 191 individuals that have lived in Germany during and after the pandemic, will provide quantitative data, allowing for the analysis of trends, patterns, and general consumer tendencies in online shopping. The survey focuses on finding the changes in online shopping frequency, product categories, spending patterns, and consumer expectations from online retailers. Furthermore a series of in-depth interviews with 10 individuals offers qualitative insights into personal experiences, perceptions, and behaviors related to online shopping during the pandemic. These interviews help in understanding the nuanced shifts in consumer attitudes and the psychological factors influencing online purchasing decisions. The personal and emotional narratives and perceptions of an interviewee makes the qualitative interviews provide an enriching and underlying understanding of the more general and broader trends of online shopping that the quantitative survey's statistical data provides. The use of a mix-method approach comes with many advantages and limitations in both the process and analysis of the data.

Empirical research suggests that online shopping activity had greatly increased during the pandemic. This resulted in the shift of consumer preferences and changes of policies. Decisions of consumers also had a great impact due to different logistical issues. Preliminary expectations of the research include that since the beginning of Covid-19 there is an increase of online shopping activity, a shift in digitalization and logistics within Germany as the standard of demand for further digital payment options and faster delivery times has increased. Potentially, an emotional detachment of in-person shopping has occurred. It is uncertain to tell whether these shifts in consumer behavior are permanent or temporary.

A holistic understanding of the effect on the German population will be drawn, with a discussion on the implications of these findings for online retailers and e-commerce policy, reflecting on the study's limitations and the need for further research in this rapidly evolving field. The findings of this study can further enable businesses and policymakers in the online shopping sphere with the possibility to adapt and prepare to current post-pandemic and future-pandemic consumer trends.

2. Introduction

The Covid-19 pandemic had greatly affected billions of people around the world, bringing along everlasting changes to human society. One drastic change could be seen within consumer shopping behavior in the online shopping sphere. In Germany, known for its robust economy and penetration in e-commerce, the pandemic was a catalyst that brought both local and international stores to new heights by the average German consumer. A specific sector of online business that has seen a great increase in sales and profits during the pandemic is Germany's online grocery retail (Dannenberg et al., 2020).

The surge in online shopping reflects a broader shift in consumer behavior, driven by the necessity of minimizing physical contact and adhering to lockdown restrictions. This rapid adoption of e-commerce platforms was not limited to grocery shopping alone but extended to various other sectors such as electronics, fashion, and home goods, demonstrating a significant shift in the shopping paradigm (Kunkel et al., 2023). The digital transformation accelerated by the pandemic has led to lasting changes in how consumers interact with retailers, prioritize convenience, and seek out digital solutions for their shopping needs.

Germany's e-commerce market, already one of the largest in Europe, experienced unprecedented growth as consumers embraced the safety and convenience of online shopping. This shift was supported by advancements in digital infrastructure, the proliferation of smartphones, and the increasing reliability of delivery services (Hardaker, 2022). Local retailers quickly adapted by enhancing their online presence, while international e-commerce giants solidified their foothold in the market.

The goal of this study is to understand the underlying reasons for changes within the e-commerce sphere of Germany during the Covid-19 pandemic and what effect it could have for both a company and a consumer in the future. By analyzing consumer behavior patterns, technological adoption, and market dynamics during this period, this research aims to provide insights into the long-term implications for the retail industry. Understanding these trends is crucial for businesses to adapt and thrive in the post-pandemic world, as well as for policymakers to support a resilient and sustainable digital economy. Furthermore, this thesis will explore the potential benefits and challenges that the shift to online shopping presents, considering factors such as digital equity, consumer privacy, and the environmental impact of increased delivery logistics.

2.1 E-Commerce market of Germany

According to the German Retail Association (Handelsverband Deutschland), the German e-commerce market has experienced strong growth over the years. In 2020, Germany's e-commerce had a gross sales of 72.2 billion euros, this further increased by 2022, in which total sales were approximately 84.5 billion euros . As of right now Germany has the largest online market in Europe and is set to further grow in the coming years with increasing technological advancements. Trends in online shopping include the allowance of product returns and same-day delivery services, highlighting the market's focus on enhancing customer convenience (Brusch & Stüber, 2014). Germany also holds high importance in the security of personal goods and data. In 2017, the European Commission's Notice on the surveillance of online products and its proposal for additional new regulations on enforcement of product compliance rules indicate the importance and focus on maintaining a high level of safety for consumers shopping online (Ullrich, 2018). Furthermore, current existing laws such as the 'General Data Protection Regulation (GDPR)' mandates how online businesses use their user's data.

Germany's strong emphasis on safety and security in its e-commerce market stems from the nation's historical focus on risk management and consumer protection in financial transactions. This cultural trait results in a cautious attitude towards online shopping and e-commerce (Sun, 2011). This German cultural aspect can be seen in working environments and industry practices, which creates a strong mutual trust between retailer and consumer.

2.2 Online shopping behavior in Germany

The safety and security in the German e-commerce market explains the massive growth it has had in recent years. Furthermore, technological advancements such as the introduction of high-speed internet infrastructure and digitalization of everyday stores has transitioned many German citizens to online shopping (Piroth et al, 2020). Popular German retail stores such as Rewe, Edeka, and Media Markt, have brought the option of online ordering and delivery, which has seen major successes in both the young and elderly, mainly due to convenience. The pandemic has changed the landscape of online shopping, and with the closing of in-person shops, many were forced to try out the online counterparts. Due to sudden popular demand during the Covid-19 pandemic, online shopping brought various problems and affected people's perception of this new shopping method, problems such as delay in delivery and sudden increases in product prices. The increase in online grocery shopping has impacted pricing dynamics

within the market. Price hikes were particularly notable for essential items, mainly driven by heightened demand and supply chain disruptions (Fedoseeva & Van Droogenbroeck, 2023).

Furthermore, the International Trade Administration (ITA) states that the most popular products that were purchased online in Germany include clothing (76% of online shoppers), shoes (73%), food & drinks (69%), consumer electronics (61%), books/movies/games (58%) and healthcare products (56%). According to the ITA, consumers have different values and reasons for purchasing a product depending on the category they are shopping in. When it comes to clothing and food industries, German consumers are said to be more price-sensitive, whereas consumers would spend a premium on products like smartphones and shoes. In addition, Germans are more likely to try out online shopping for the first time due to their trust and confidence in their system (Sun, 2011). Part of the trust in their system stems from the payment methods that are available for online shoppers. In Germany, the most common methods of purchase are on invoice, PayPal, direct debit and credit cards. Furthermore, the option of 'Buy now, Pay later' has influenced many individuals to purchase expensive goods that are considered unaffordable, especially among the younger generations. This has greatly benefited onlines businesses but drawbacks for some consumers can be seen. Additionally, with the 14-day return policy that allows the refund of any online purchases within 14 days despite being defect free, allows consumers to try out products and refund it if they simply do not like it. The 14-day return policy is a major factor for many to try out and continue being part of the online shopping experience, as it allows for risk free trial and error.

Despite the acceleration and improvements made to online grocery shopping in Germany, it is said that it still lags behind its international counterparts. Nearly half of the German population is aged over 50 and the 55-64 age group is emerging as one of the largest user segments in e-commerce (Braun & Osman, 2024). Elderly in Germany are commonly overlooked as a potential market, but with growing health problems and lack of nearby grocery stores, the elderly are becoming an attractive target market for online grocery stores.

2.3 Changes of businesses during the pandemic

Looking at the pandemic from a business standpoint, a lot of companies at the beginning of the Covid-19 pandemic faced extreme hardships and even bankruptcy. According to a survey conducted on nearly 5,800 different small businesses, the majority of the companies only had 2 weeks worth of cash on hand, highlighting their vulnerability and reasons for many of their downfalls (Bartik et al, 2020). At the beginning of the pandemic, 66% of medium-sized enterprises (SMEs) suffered turnover losses, however by September 2021 many businesses had recovered and found a solution bringing this statistic to 28% (Fritzi and Volker, 2023). Furthermore, the introduction of lockdowns also affected companies, and different containment measures helped against problems such as worker absences and supply chain disruptions. To both cut losses and follow new temporary regulations in which customers were unable to shop in-person, companies had to adapt. Supply chain issues were solved through adaptations and strategies such as intertwining, scalability, substitution, and repurposing (Ivanov, 2020). Multiple businesses went online and created a digital platform in which customers could shop through simple and quick means. Businesses that had transitioned from an offline platform to an online platform were able to cut losses, while companies that had already established a digital business prior to the pandemic had seen tremendous increases in profits, most notably, Amazon and Netflix. According to Amazon's annual report, between the time period of January, 2020 and December, 2023, Amazon had seen a total increase of 142.63% in net income. Through further analysis of their earnings release and by investigating other factors such as revenue, sales and news of expansion, it can be concluded that Amazon had come out of the Covid-19 pandemic with great growth of their company, despite having some difficulties in the year of 2021 and adjusting to the drastic changes of the lockdowns. According to the International Trade Administration (ITA), in 2021, Germany's most popular online retailers were amazon.de, otto.de, zalando.de, mediamarkt.de and ikea.com. Despite the ranking making the competition look close, Amazon, the only non-European company, has seen more net sales in the fiscal year of 2021 than all the other previously listed competition combined. It is debatable whether the market leader Amazon, a foreign company in Germany, is beneficial for the German e-commerce market. Despite their conformity to local laws, in 2023, a lot of backlash was received by their own employees and multiple worker strikes were conducted.

Overall, the e-commerce market was one of the very few sectors that was able to withstand the impact of the Covid-19 pandemic. Issues were mostly temporary and lasted for only a short time, in comparison to sectors like the hospitality industry in which consequences can still be seen. Thus it can clearly be said that the e-commerce market was adaptable and rebounded well to overcome the challenges brought upon

by the pandemic. Despite this, new onlines stores had to quickly adapt with the use of different strategies. To compete with competition initially, prices were reduced, discounts were given, social media marketing reached new records in expenditure, and most notably retaining existing customers through loyalty programs and personalized offers were more cost-effective than acquiring new ones. Focusing on customer satisfaction and adapting services to meet changing needs was essential (Mastari, 2020). With a new and large online shopping environment, many companies had spent new records on analytics of the market and customers. Customer analytics can include customer behavior, preferences, and buying patterns (Davenport & Harris, 2007). Understanding how the market was to change and customers behaved could give companies an edge. Predictive analytics could provide a business with better preparation for forecasted demands and inventory management (Fosso Wamba et al., 2015). This was even more reason as to why smaller businesses that had no online presence faltered early on as marketing cost more than ever to gain new customers and customer retention was near impossible as the foundation was not built around their company yet.

3. Research introduction

The following research will use a mixed-method approach. Firstly, a survey will be conducted at different locations within Germany and secondly, an in-depth interview will be done with different German citizens to further understand the reasoning behind their survey choices.

Surveys are a pivotal component of this research as they provide a broad quantitative understanding of consumer behaviors, preferences, and trends. By administering a structured questionnaire to a large sample of participants, this study can statistically analyze consumer behavior on a large scale, ensuring that the results are generalizable across the broader population. Surveys are particularly effective in capturing data that can be quantified and analyzed to reveal patterns and correlations of a general target population (Almeida, 2018), such as the frequency of online purchases, spending habits, and preference for certain product categories during the pandemic. Analyzing the product categories can have a more detailed view of what markets in the e-commerce sphere were affected the most.

Surveys provide data from a surface level and are good to get a general understanding of certain demographics, however, interviews provide depth and delve into the reasonings to the data and answers that were given in the survey. Qualitative interviews with a select group of consumers allow for an exploration of complex thoughts, emotions, and motivations that influence shopping behavior (Clark, 2017). These interviews enable the researcher to delve into detailed consumer experiences that are not

easily captured through surveys. By incorporating open-ended questions, interviews facilitate a deeper understanding of the psychological and social factors within online shopping behavior.

Due to conducting the surveys and interviews in both English and German, it was decided that a non-structured method would be implemented in which clarification of questions can be conducted for participants in need of help. Furthermore, certain conditions such as environment, time, date and place differed, and certain data was collected under different conditions.

4. Methodology

4.1 Quantitative research design

Table 1. Survey location information

	Location	**Date**	**Time**	**Weather**	**No. of responses received**
1	Bensheim, Town Square	Saturday, 27th of April	14:00-17:00	Sunny	34
2	Frankfurt am Main, Zeil	Thursday, 2nd of May	11:00-14:00	Sunny	60
3	Heidelberg, Altstadt	Sunday, 5th of May	16:00-17:00	Sunny	19
4	Online/Others	Tuesday, 7th of May	-	-	78

Survey locations were chosen based on their daily activity to receive as many responses as possible. However, due to these locations being very high activity, places like Frankfurt, Zeil and Heidelberg, Altstadt can be seen as very touristic locations and could have had an effect on the validity and authenticity of the data that was collected. Dates and times were planned to be done under sunny weather conditions to ensure responses were not done in a rush and more responses could be collected.

Firstly, Bensheim was chosen as a survey location for being a town that is very low in population and has a much older average population compared to Frankfurt and Heidelberg. According to the city Bensheim's 2018 demographic report, out of a population of 41,562, the average age of the town is 45 years old, with the median being 46 years old.

Frankfurt was chosen as the second location due to it being known as one of the most active and advanced cities of Germany. Despite being a financial hub for international companies (Grote, 2002), it also sees a lot of travel from the rest of Germany whether it is short-term or long-term. However, since the survey was conducted during a work day and within working hours, the data collected may be a bit biased and could not have incorporated enough data to represent the average working class.

Heidelberg was the final location due to it being a middle ground between Bensheim and Frankfurt. Despite having a very large population of students, it also has a large community of elders with active tourism. A level of existing tourism within the city is important as it usually indicates a city of better infrastructure, wellbeing and growing community. Tourism also indicates that the city is built more towards consumption (Mullin, 1991).

Finally, surveys were also sent out and posted on online German communities such as the popular online forum 'Reddit'. Furthermore, it was advised that individuals who have taken the survey, raise awareness by word of mouth and make it known to either friends and families. However, it is not known to what extent word of mouth had helped in collecting further data as it had not been measured. Online surveys in general help in collecting a large amount of data in a short time. It is important to note that the more sensitive and private the survey questions are, the more likely it is to be more accurate (Murdoch et al., 2014).

Table 2. Survey questions

	Question	**Possible Answers**	**Reason of choosing question**
1	Did you primarily live in Germany during the Covid-19 pandemic?	- Yes - No	To separate relevant answers and get the correct target audience of the research. If no was answered, no further questions were asked. If yes was answered, the survey would begin.

2	What is your gender?	- Male - Female - Prefer not to say - Other	To see the differences and trend lines of different demographics of the target population. Whether gender has any effect within consumer behaviors related to the online sphere.
3	What is your age group?	- Under 18 - 18-24 - 25-34 - 35-44 - 45-54 - 55-64 - 65+	To determine the differences and trend lines of different demographics of the target population. Whether age has any effect within consumer behaviors related to the online sphere.
4	How frequently did you shop online before the COVID-19 pandemic?	- Daily - Weekly - Monthly - Rarely - Never	To get an understanding and a comparison point of shopping frequency for this surveyee. Using this answer and comparing it to the following questions can lead to a deeper understanding of shopping frequency, changes and potential patterns.
5	Since the beginning of the COVID-19 pandemic, how has your overall frequency of online shopping changed?	- Increased significantly - Increased slightly - Remained the same - Decreased slightly - Decreased significantly	This answer will provide a second point of comparison for question 4. This can provide a first impression of the participant and indicate how their purchasing has changed.
6	Which product categories have you shopped for	- Electronics - Groceries - Clothing	This question, along with the following, can determine the markets that were most affected by the

	online BEFORE the pandemic?	- Pharmaceuticals - Home and Garden - Entertainment - Food - None - Others	pandemic and look at an individual's change in preferences.
7	Which product categories have you shopped for online DURING the pandemic?	- Electronics - Groceries - Clothing - Pharmaceuticals - Home and Garden - Entertainment - Food - None - Others	This question, along with the previous, can determine the markets that were most affected by the pandemic and look at an individual's change in preferences.
8	What are the main reasons for any increase in your online shopping activity during the pandemic?	- Convenience - Safety concerns - More free time - Limited access to physical stores - Price advantages - Others	This will obtain the reasons as to why certain markets saw an increase in sales. It can also gain the underlying reasons as to why other markets failed and which aspects they failed to adapt in.
9	How has your average monthly spending on online shopping changed from before to during the COVID-19 pandemic?	- Increased significantly - Increased slightly - Remained the same - Decreased slightly - Decreased significantly	Investigates the public's changes in expenditure behaviors and how prices have changed for certain markets. Along with other answers, this question can provide an understanding of how online shopping differs from in-person shopping from a consumer perspective.

10	Do you prefer shopping from local German online stores or international online stores?	- Local - International - No preference	Takes into consideration whether the pandemic and not having the possibility to go to stores in-person had affected either local or international stores negatively.
11	After the pandemic, do you intend to continue shopping online?	- Yes, more than now - Yes, the same as now - Yes, less than now - No, will shop in-person	This will provide an insight into the future and how online businesses will most likely be affected once shopping conditions return to normal.

4.2 Qualitative research design

Table 3. Interview participant information

	Location	**Gender**	**Age**
1	Bensheim	Male	45-64
2	Bensheim	Female	18-24
3	Frankfurt	Female	55-64
4	Frankfurt	Female	18-24
5	Heidelberg	Male	35-44
6	Heidelberg	Female	18-24

Table 4. Interview questions

	Question	Reason of choosing question
1	Which online store/company have you bought the most from during the pandemic?	To specifically identify which market/company had the greatest impact on an individual and what strategies were used in gaining a customer during the pandemic.
2	What motivates you to choose online shopping over in-store shopping? Are there specific events or experiences that influenced this preference?	To discover the motivation behind online shopping and what specifically triggered the motivation.
3	How do you assess the trustworthiness and safety of an online store? Have you ever had an experience that made you more cautious about online shopping?	To gain an understanding of how individuals adapted to a new shopping environment and how negatives were avoided such as scams end etc.
4	How has the pandemic specifically affected your feelings or attitudes towards online shopping?	To find out the direct link between the pandemic and online shopping within a consumer.
5	How do you see your shopping habits evolving in the next few years? Do you think the changes in your shopping behavior during the pandemic are temporary or permanent?	This question will examine how an individual will adapt to the future and how they have changed from before to after the pandemic.
6	What does an online store have to do to gain your loyalty? Can you describe an online shopping experience that either won your loyalty or drove you away?	This will analyze and gain an understanding of how customer retention worked during the pandemic. Furthermore, it will indicate what strategies were used to come out on top for their company. This could include weekly newsletters, reminders, discount coupons or a loyalty system.

7	How does online advertising and marketing affect your shopping behavior? Are there specific ads or campaigns that have resonated with you deeply?	This question takes another look at the marketing of businesses during the pandemic and tries to specifically narrow down on the advertisements that were used. Furthermore, this will delve further into emotion and how an individual's perception is affected.
8	After making an online purchase, have you ever felt regret or satisfaction? Can you describe what influenced those feelings?	If the previous question was to understand the emotions before a purchase, this question looks at the emotions after the purchase was made. This looks at the user experience, interface and other factors that could influence an individual's choice. Through this a connection can be made whether the pandemic influenced potential improvements and whether this could affect in-person businesses in the future.

Interviews were conducted after the survey and only a select few were selected to further question and gain an understanding of their survey answers. To avoid selection bias, random participants were selected at random intervals, two were picked at each location, Bensheim, Frankfurt, and Heidelberg.

5. Results

5.1 Quantitative results

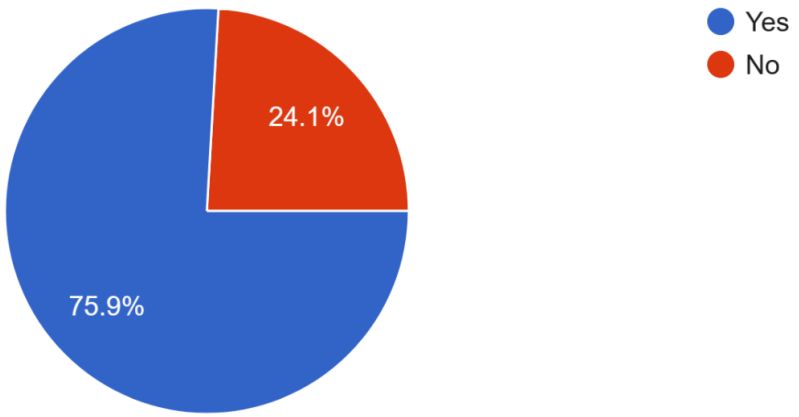

Figure 1. Percentage of surveyees that lived primarily in Germany during the Covid-19 pandemic

A total of 191 results were collected, of which 75.9% is relevant for this research. There is a high likelihood that due to having conducted the surveys on the streets of very populous and tourist heavy areas, that a large proportion of them had not lived in Germany during the Covid-19 pandemic. The individuals that had answered 'No' were not further asked and the remaining 145 participant's data was collected and further analyzed.

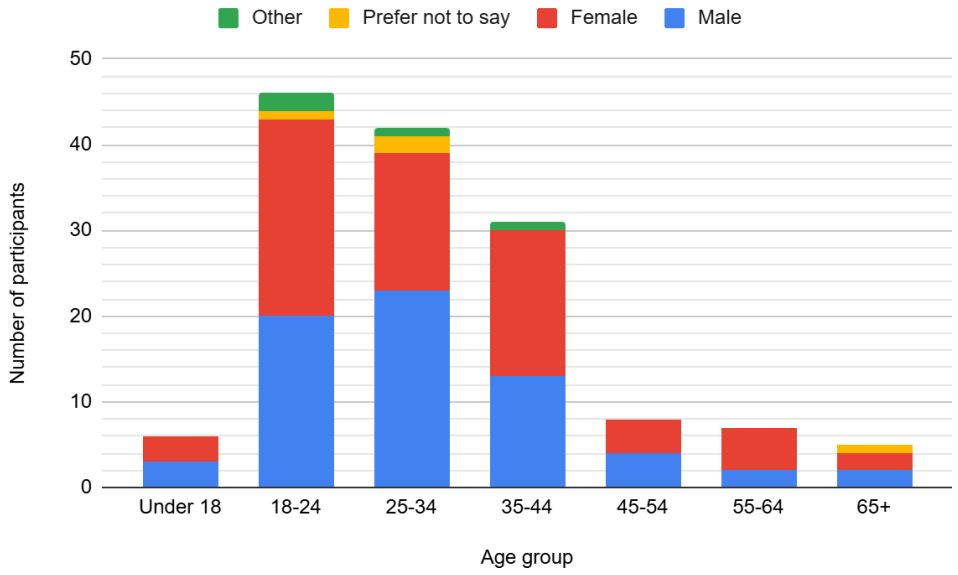

Figure 2. Population distribution by age

The participants' age group was split into seven different categories. The majority of surveyees were between the ages of 18 and 44 (119 out of 145) which could be related due to the locations and conditions the survey was conducted in. Furthermore, the gender distribution of the target population was split nearly evenly between 'Male' and 'Female', in which the male population accounted for 46.2% and the female population 48.3%. Unfortunately, due to the very low sample size of 'Other' and 'Prefer not to say', the data and results that are extrapolated have very low validity and reliability. Thus, no accurate conclusions can be made for either population. The results and conclusions drawn from this survey will mainly represent the younger generations of Germany. Despite having a lower population of elders represented, the information gathered will still be a valuable asset for both companies, consumers and businesses for older population areas of Germany.

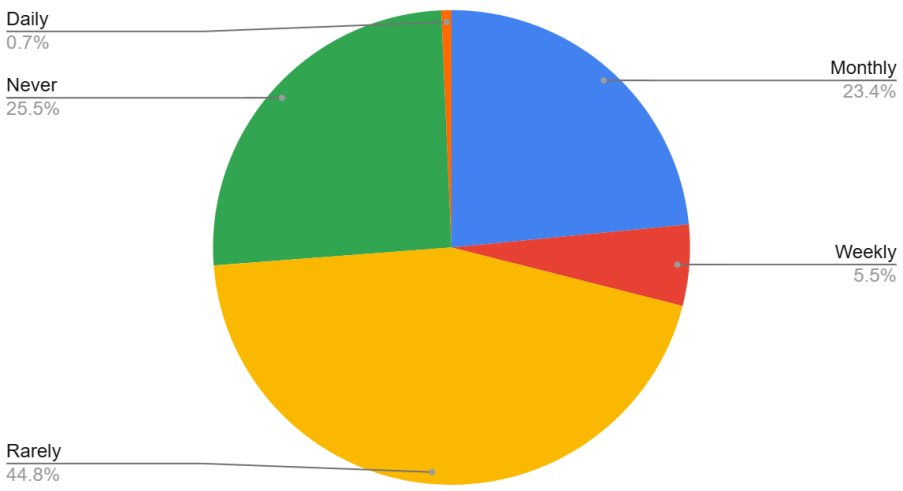

Figure 3. Online shopping frequency before the Covid-19 pandemic

The majority of survey results show that before the Covid-19 pandemic nearly 75% had either 'Never' or 'Rarely' shopped online. Despite this pie chart showing that a large portion had not shopped online before, the opposite argument can be made that a significant portion of answers shows that they shop at least once a month or more. The results showcased in figure 3, will be important when compared to the following question "Since the beginning of the COVID-19 pandemic, how has your overall frequency of online shopping changed?" as it will show which individuals were impacted more.

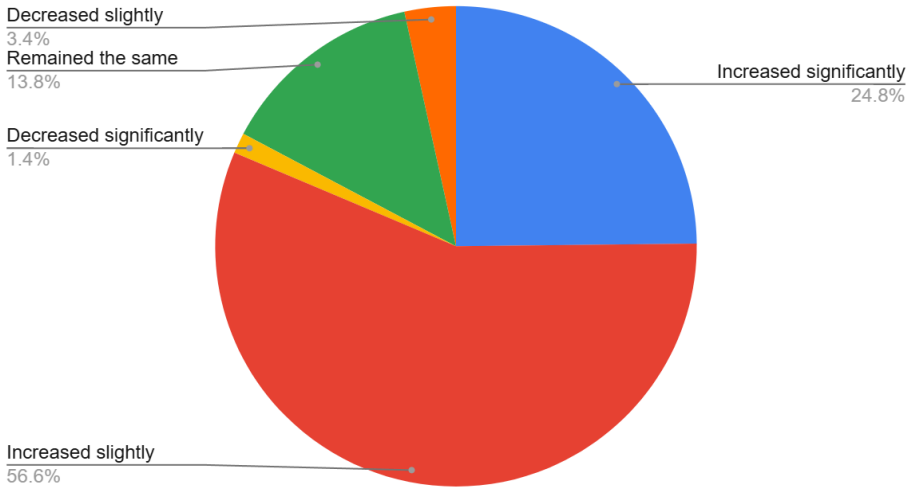

Figure 4. The change in frequency of online shopping after the Covid-19 pandemic

The results of this question are staggering and an increase in online shopping activity either slightly or significantly can be seen within 81.4% of participants. Despite these one-sided results, a decrease in only shopping was present in a few individuals which is surprising as lockdowns should have motivated the majority of shoppers to go online more frequently. However, it depends on which answer groups eventually increased and/or decreased. Additionally, no clear connections could be seen between the results and demographics of people surveyed.

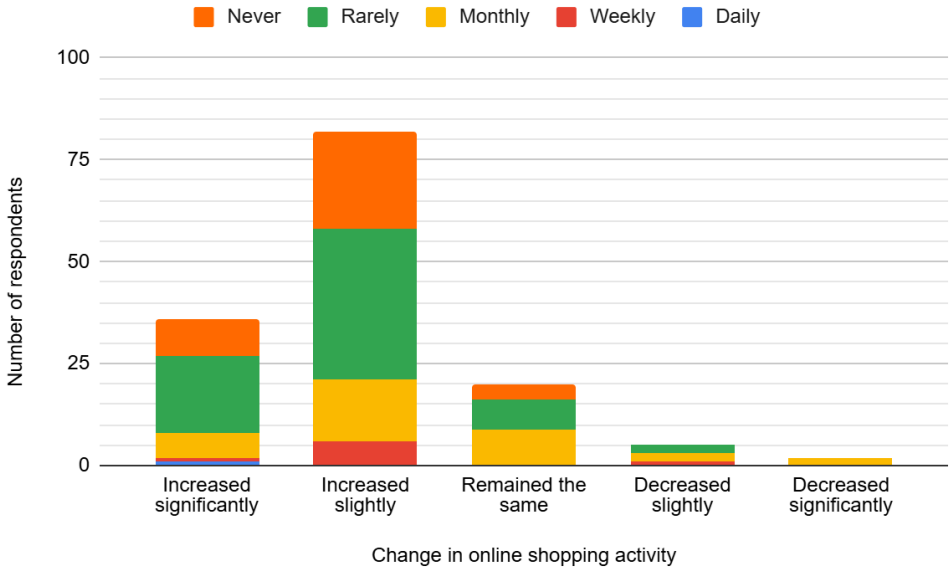

Figure 5. The change in online shopping frequency from before to after the pandemic

When combining the data of the two questions, 'How frequently did you shop online before the COVID-19 pandemic?' and 'Since the beginning of the COVID-19 pandemic, how has your overall frequency of online shopping changed?', the conclusions and relationships between the questions can clearly be identified. Overall, it is apparent that most individuals answered either 'Increased significantly', 'Increased slightly' or 'Remained the same' and all of the participants that have selected 'Never' were all categorized within either an increased or equal state of shopping frequency. The data of figure 5 suggests that the pandemic acted as a catalyst for increasing online shopping activity among the majority of citizens in Germany. The lockdowns and social distancing measures necessitated a shift to online shopping for a variety of goods, including essentials that were previously purchased in physical stores. In summary, whether individuals had tried online shopping before or had some prior knowledge and activity before does not seem to matter; the lockdowns forced individuals to partake in online shopping at least a handful of times. The data reflects that the pandemic has not only increased the

frequency of online shopping across different demographics but also expanded the base of online shoppers, integrating online shopping more deeply into daily consumer habits.

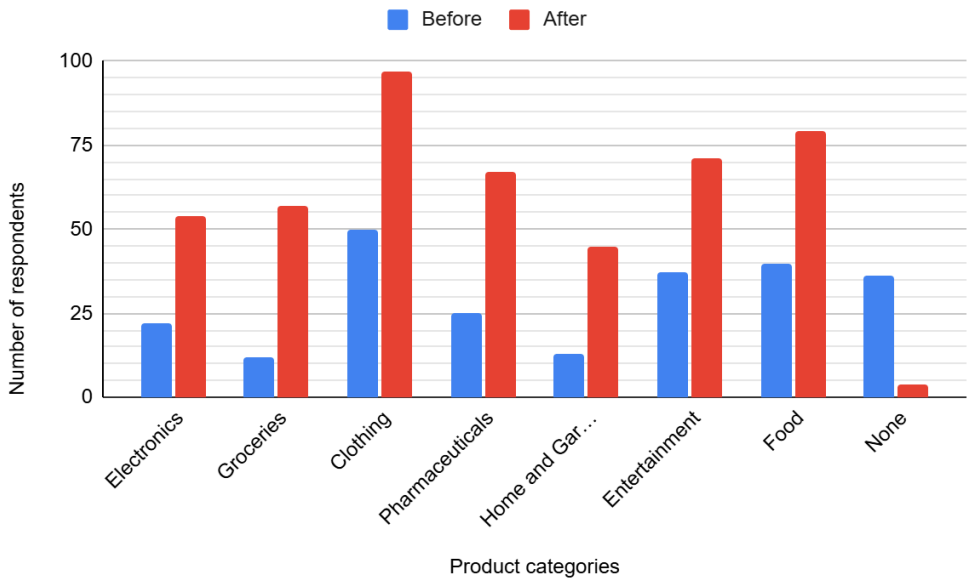

Figure 6. Comparison of product categories that were shopped online before and after the pandemic

Through a comparison between the product categories that were purchased before and after the pandemic, it can be concluded that every single category has experienced a significant increase in online purchase rates. The smallest increase was observed in the entertainment sector, with a notable 91.9% rise, indicating a substantial shift despite being the least affected category. The largest increase was seen in the home and garden sector, which experienced a staggering 246.15% increase in purchase rate since the pandemic started.

Interestingly, it can also be seen that the number of respondents selecting the 'None' option for online purchases has decreased dramatically. This showcases that nearly all individuals had begun or at least tried shopping online once the pandemic ensued. The sharp decline in the 'None' responses highlights a critical shift: individuals who previously refrained from online shopping were compelled to adopt this method due to pandemic-related restrictions and safety concerns.

These results could be a reflection of the real-world economic landscape, indicating which business sectors prospered the most and which faced challenges during the pandemic.

The largest increase in the home and garden category suggests that many people invested in home improvement and gardening projects, likely due to spending more time at home. This trend reflects a broader shift towards enhancing living spaces and engaging in home-based activities.

Although the entertainment category saw the smallest increase, a 91.9% rise is still significant. This may indicate that while people sought new forms of entertainment, many also utilized existing subscriptions and platforms rather than purchasing new products.

Categories like electronics, clothing, and groceries likely saw considerable increases as well, driven by the need for home office setups, comfortable loungewear, and contactless grocery shopping.

The widespread adoption of online shopping, as indicated by the decrease in 'None' responses, implies that nearly all surveyed individuals have become familiar with and have utilized online shopping platforms. This trend suggests a permanent shift in consumer behavior, where the convenience and safety of online shopping have been widely recognized and embraced.

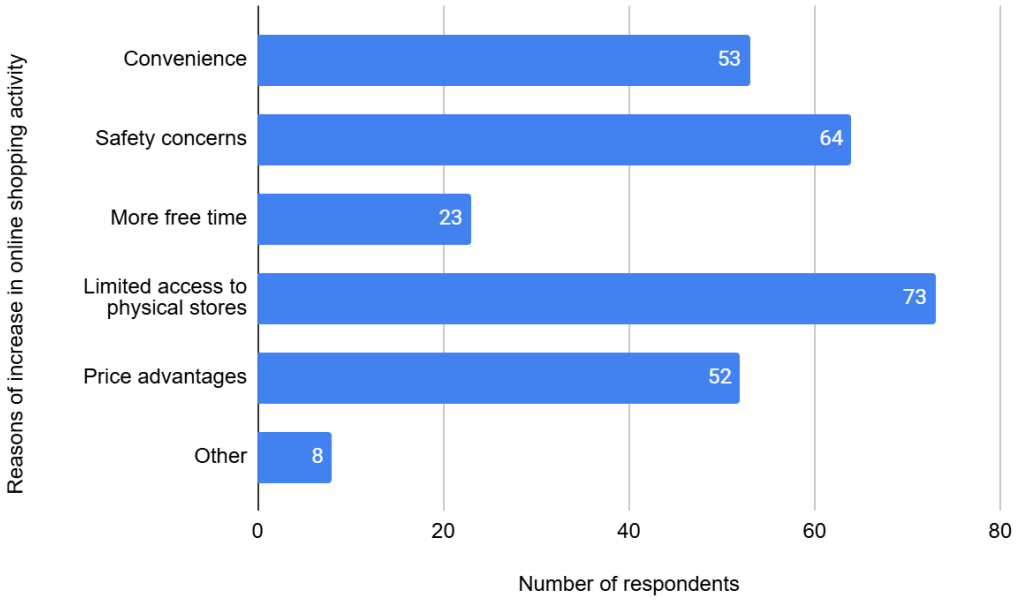

Figure 7. Reasons of increase in online shopping activity during the pandemic. 'Other' includes individuals that had not seen increased online shopping activity.

When asked about the reasons for the increase in the participants' online shopping activity during the pandemic, the majority had agreed upon the fact that less physical stores were available or were

inaccessible. Also, results clearly demonstrate that convenience, safety concerns, limited access to physical stores and price advantages are the majority and the main reason for increased activity. Surprisingly, free time did not seem to play a significant factor in dictating online shopping activity, this can indicate that the increase in popularity of online shopping was not a choice but rather a result of lockdown regulations.

Table 5. The change in average monthly spending on online shopping from before to after the Covid-19 pandemic

Change in monthly expenditure	**Number of respondents**
Increased significantly	36
Increased slightly	86
Has not changed	19
Decreased slightly	3
Decreased significantly	1

The data on changes in average monthly spending on online shopping before and after the pandemic reveals significant shifts in consumer behavior. Out of the total respondents, a majority observed an increase in their online shopping expenditures, with 36 respondents indicating a significant increase and 86 reporting a slight increase. Only a small fraction of the respondents, specifically 19, noted no change in their spending habits. Conversely, a minimal number of participants experienced a reduction in their online shopping, with 3 respondents reporting a slight decrease and just 1 respondent indicating a significant decrease. This data underscores a prevalent trend of increased online shopping activity post-pandemic.

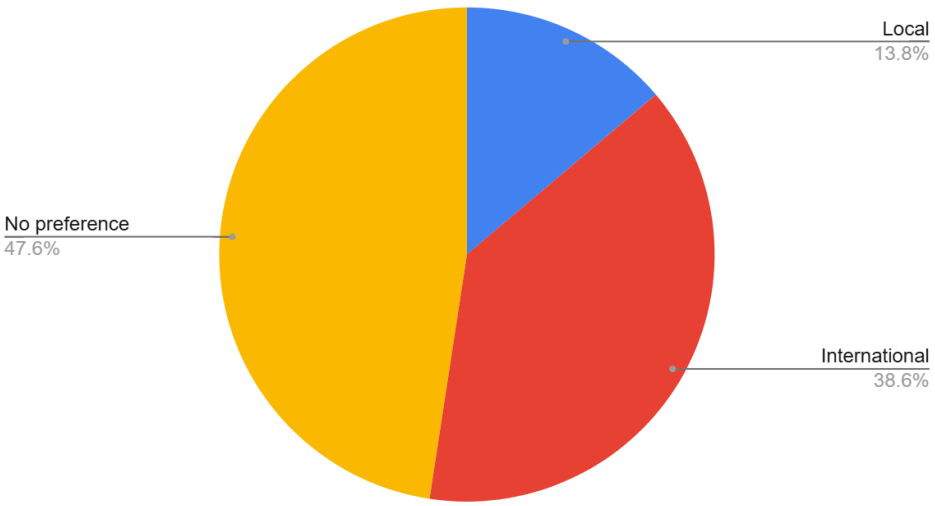

Figure 8. Preference of shopping at local or international online stores

Upon analyzing the survey results, it was determined that the majority did not have a preference between shopping at local or international online stores. However, despite the majority having no preference, a large proportion of respondents prefer international stores over local stores. This can have a large impact on local businesses and provide an explanation as to why many smaller stores were unable to survive the covid-19 lockdowns. Furthermore, even with 47.6% of individuals having no preference, it is very likely that in a neutral internet usage, one will be more exposed to a large international corporation's marketing than a small local business, without having the intent to. Additionally, a high preference for a large international store can be harmful for local businesses, but beneficial and more convenient for an individual with product availability, costs, delivery times and more.

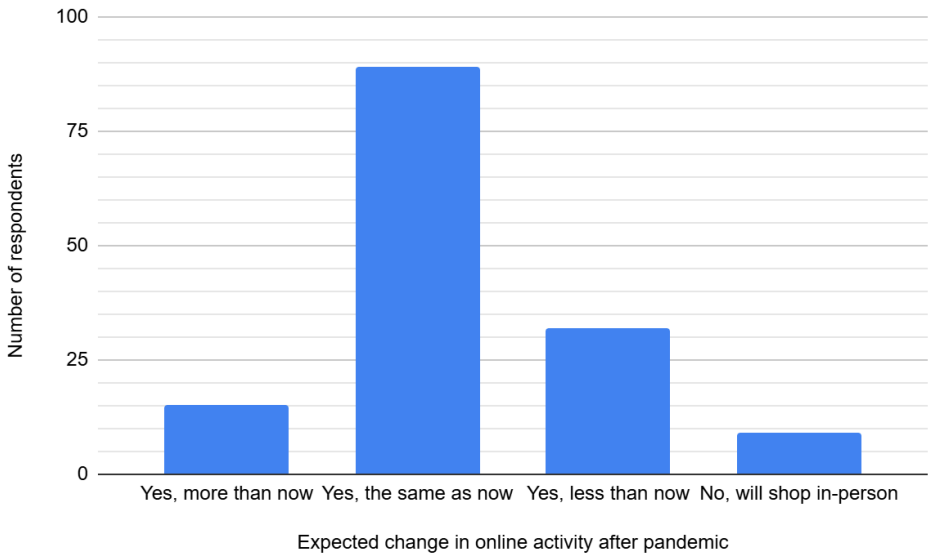

Figure 9. The amount of online shopping intended after the Covid-19 pandemic

Finally, the pandemic has forced the majority to spend more than they usually do on online shopping. Despite realizing the increased expenditure on online goods, when asked about whether the surveyees would intend to continue shopping online after the pandemic, over half had answered that they intend to continue shopping the same amount as during the pandemic. It is interesting to see that a significant proportion plans on shopping more than they did during the pandemic, but this community mainly consists of individuals that had little to no experience about online shopping before the pandemic. Overall these results indicate that their experience of online shopping was at least satisfactory and that they would further continue shopping online. The individuals that are planning on reducing their online shopping activity indicates that they were somewhat forced into it by lockdown regulations or that they had negative experiences.

5.2 Qualitative Results

Table 6. Interview response summary and observations

	Question	Observations
1	Which online store/company have you bought the most from during the pandemic?	Among the interviewees the most commonly mentioned was either Amazon or Netflix. Occasionally grocery stores or electronic stores were mentioned as well, along with other streaming services such as Prime (Amazon) or Disney+. Despite living in Germany, most of the companies that were mentioned were large companies from the United States.
2	What motivates you to choose online shopping over in-store shopping? Are there specific events or experiences that influenced this preference?	Every single one of these participants answered 'Convenience' in their survey but upon further questioning it narrowed down to lack of options or simply lockdown regulations. Interestingly, participant 1 mentioned that due to the feeling of boredom, purchases were often made either on Amazon or different clothing brands. Despite answering the first part of the question, all the other participants could not relate it to either a feeling or specific event that has resulted in them purchasing items online.
3	How do you assess the trustworthiness and safety of an online store? Have you ever had an experience that made you more cautious about online shopping?	Most of the interviewees did not give an answer nor did they know how to assess the trustworthiness of an online store. If anything, the individuals were blindly trusting that the stores were safe. However, participant 4 stated that once when purchasing jewelry online from an international store, a fake product was delivered from China. According to her, she had blindly believed that the website was real and was rushed by an Instagram ad with a discount. The discount made her hastily overlook the details and since then she takes her

		time shopping to inspect the links, websites or other details.
4	How has the pandemic specifically affected your feelings or attitudes towards online shopping?	All the individuals were satisfied with shopping online but many mentioned that it is starting to turn them away as costs were drastically increasing when it comes to either food delivery, entertainment/streaming services or the quality of products has dropped. Additionally, for delivery of groceries, it was mentioned that many times the fresh produce sent is not as good as if one were to select them in-person. According to one participant, it could be seen that quality often suffered over time, and it could be seen that shops would try and cut costs wherever possible.
5	How do you see your shopping habits evolving in the next few years? Do you think the changes in your shopping behavior during the pandemic are temporary or permanent?	A few individuals mentioned that adjusting the expenditure is very important since many of them were unable to keep track of their money when going on a purchasing spree online. To add on, many also mentioned that they want to reduce their time spent online as it gets them disconnected from human interaction with either a store employee or advisor. However, all of the participants stated that their pandemic behaviors were temporary and that many have already started adjusting to their 'goals'. Also, two individuals mentioned that due to the lack of stores available in the city, that many will avoid shopping in-person as there are more options online and products can be compared with online reviews.
6	What does an online store have to do to gain your loyalty? Can you describe an online shopping experience that either won your	Interviewees stated that discount or loyalty points are an important aspect of keeping their loyalty. Furthermore, participant 6 stated that constant exposure to advertisements and regular email newsletters

	loyalty or drove you away?	convinces them to visit their store and/or purchase something. To the question of what has won or drove away an online shop's loyalty, many mentioned that price increases have made them stop certain services or made them completely lose interest. Although many were driven away from a store, all the individuals mentioned that their loyalty was won through their own choice and none of it was orchestrated or planned by the businesses.
7	How does online advertising and marketing affect your shopping behavior? Are there specific ads or campaigns that have resonated with you deeply?	When mentioned about online advertising or marketing many have mentioned an experience that has influenced them to at least visit their site. Participant 2 mentioned that a product with eco-friendly properties usually persuades them more than normal counterparts. Participant 5 has also mentioned that a discount on an electronic product has convinced him to purchase it but eventually returned the product due to it not being what he expected.
8	After making an online purchase, have you ever felt regret or satisfaction? Can you describe what influenced those feelings?	Most participants felt satisfaction with their online purchases when products met expectations, were well-reviewed, and arrived on time. However, feelings of regret often stemmed from misleading product descriptions, poor quality, and complicated return processes. Influences on these feelings included persuasive marketing, positive reviews, and influencer endorsements. Overall, most participants when thinking back to their purchasing during the pandemic seemed to have positive memories of online shopping.

6. Implications

The primary research that was conducted to gain both quantitative and qualitative data showcases different correlations and brings a holistic understanding of how the Covid-19 pandemic had affected people's online shopping behavior. Both businesses and consumers can use the findings to further improve themselves and gain an understanding of the online shopping landscape of Germany.

The surge in online shopping behavior since the pandemic presents a significant opportunity for businesses to refine and enhance their digital strategies. Firstly, businesses can leverage this data to optimize their e-commerce platforms, ensuring they are user-friendly, efficient, and equipped with the latest technology, this includes their user interfaces, logistics or even customer support. According to survey results, convenience plays a major role in a customer's reason for shopping online, enhancing the user experience through intuitive navigation and personalized recommendations can attract and retain customers who have newly transitioned to online shopping.

Secondly, businesses can utilize this data to improve marketing strategies. By targeting advertisements based on consumer behavior analytics, businesses can reach a broader audience and tailor their messages to meet the specific needs of online shoppers. This targeted approach can result in higher engagement rates and conversion rates, driving more sales through online channels. According to the qualitative collected, advertisement seemed to have played a major role in attracting customers and keeping loyalty to those already existing.

Moreover, the data on increased online shopping can help businesses expand their product offerings and diversify their inventory. By analyzing which products are most frequently purchased online, companies can identify new trends and customer preferences, allowing them to adjust their stock accordingly. The findings of this research clearly showed which product categories had the highest increase and were among the most popular when individuals shopped online during the pandemic. Figure 6 showcases that the home and garden sector experienced the largest growth. Keeping this in mind, either a new product line or improvements can be made to already existing products. Additionally, businesses can explore the potential for new product lines that cater specifically to online shoppers, further boosting their competitive edge in the digital marketplace. Smaller local businesses however, lacked a quick adaptation method. For them to keep their businesses up and running, a huge potential exists in going online and it could benefit them in the long run. Through the data collected, marketing strategies can also be altered and developed to cater towards a population that is still new to online shopping, as the number seemed to

have drastically increased during the pandemic. According to the interview, marketing strategies had an effect on all the interviewees and many have different preferences. Understanding these preferences and doing a thorough market forecast or customer analysis can help businesses both prepare for upcoming trends and capitalize on existing opportunities. In connection to new marketing strategies, user experiences may have to be improved to support new individuals and not overwhelm them with too much information at once.

Furthermore the rapid evolution of Germany's infrastructure also helps consumers and businesses adjust to this new shopping landscape. Improvements in infrastructure can benefit internet connections and wider spread connectivity bandwidths enabling consumers to have a better and more available experience. Companies operating online can also benefit from improved logistics and delivery networks, with enhanced delivery times and qualities. Upgraded payment infrastructure in Germany, including the integration of multiple secure payment gateways, ensures that transactions are processed quickly and safely. Consumers can benefit from a variety of payment options, such as credit cards, digital wallets, and buy-now-pay-later services, making the purchasing experience more flexible and reliable.

On the other hand, the data and findings drawn from the survey and interview can also be beneficial for consumers. One major benefit is the ability to make better-informed purchasing decisions. With more people shopping online, the volume of product reviews and ratings has increased, along with a more technologically literate online shopping community, reducing the amount of scams and frauds that people fall for. Additionally, shoppers can avoid poor-quality products and enjoy a more satisfying and safe shopping experience. A major reason as to why individuals fall for scams in the first place is due to different factors that the sites impose on you such as time constraint, by becoming aware of these scam methods, a lot of victims can be avoided. Unfortunately, despite knowing this, fraudulent activities will continue to exist and target the vulnerable of the internet, building a support system to avoid being taken advantage of can help in reducing the amount and supporting new online users avoid being deterred from online shopping.

Another advantage lies in the competitive landscape of online retail, which has intensified due to the rise in online shopping as identified in the qualitative and quantitative results. Retailers are now more inclined to offer attractive promotions and discounts to capture consumer attention. This approach not only maximizes savings but also allows consumers to stretch their budgets further. As competition rises the amount of options will also increase. Consumers have access to a wider variety of items, including those

that may not be available in local stores. This greater accessibility allows consumers to explore unique or niche products that better match their preferences and needs.

The existence of the 2-week refund policy in Germany allows first time customers to try out a certain product and see if it satisfies their needs. According to the interviews, having the option of trial and error can help an individual eventually get the product they expect.

However, despite the increase in options available for a consumer, a majority of these products offered come from the international online market. One of the most immediate effects is the decline in foot traffic to physical stores. As more consumers opt for the convenience of online shopping, fewer people visit stores in-person. This can have a major effect on not only the small local business but even the city as a whole. Another significant issue for local businesses is the increased competition from online retailers. E-commerce giants like Amazon and other major online platforms can offer a wider variety of products, often at lower prices due to their ability to buy in large amounts. Price competition is a big problem for smaller businesses as they typically do not have the budget to gain price advantages when cooperating with suppliers. A drastic influx of online shoppers does not only have positives but also major problems. Additionally, e-commerce giants also will try to keep you trapped in their ecosystem which can often be seen with examples in Amazon, Apple and Google. Subscription models and loyalty programs further entrench customers within a brand's ecosystem. Amazon Prime, for example, provides benefits such as free shipping, streaming services, and exclusive deals, making the subscription highly valuable and giving less reason to move away from an already running subscription. Bundling services at a discounted rate, as seen with Microsoft’s Office 365, also encourages customers to stay within the ecosystem to maximize their value, and instead of purchasing one product, multiple are purchased at once. In a sense, a monopoly is created within an individual's choices of shopping, where once a person is in an ecosystem, it makes little sense to leave it due to how beneficial the offers are.

In summary, the data on increased online shopping behavior since the pandemic offers consumers a multitude of benefits. From making informed purchasing decisions and taking advantage of discounts to understanding improved customer service policies and exploring a broader range of products, consumers can navigate the online shopping landscape more effectively. Despite the multitude of benefits, a lot of downsides could also be seen as a result of increased online shopping activity in Germany.

7. Conclusion

In conclusion, the findings of this research offer valuable insights for both businesses and consumers. The results derived from the survey and interviews underscore the profound impact of the Covid-19 pandemic on the shopping behaviors of the average German. The pandemic instigated a significant shift towards online shopping, with many consumers adapting to this mode out of necessity during the lockdowns.

A key takeaway from the research is the intention of the majority of respondents to continue using online shopping platforms, albeit with a slight reduction in their expenditure. This trend indicates a sustained preference for the convenience and safety that online shopping offers, even as pandemic restrictions have eased.

From the consumer's perspective, the pandemic triggered a surge in activity across various online shopping sectors. Established businesses with robust online platforms thrived during this period, capitalizing on the increased demand. In contrast, smaller businesses faced challenges but showed resilience by many attempting to adapt to the new market conditions.

The literature review had further supported the qualitative research findings, highlighting critical factors influencing online shopping behaviors among Germans. Trust and security in the legal system emerged as pivotal reasons for the confidence in online transactions. Lockdown restrictions had forced many newcomers to try out online shopping. Despite initial hesitation, these newcomers adapted well, often finding the experience convenient and satisfactory, leading to increased expenditure.

This research provides a comprehensive understanding of the online shopping landscape in Germany during and post-pandemic. For businesses, these insights are crucial for strategizing and enhancing their online presence to cater to the evolving preferences of consumers. Large enterprises can further solidify their market position, while smaller businesses can identify opportunities to improve and innovate their online offerings.

For consumers, the study underscores the importance of making informed decisions based on the evolving dynamics of online shopping. It highlights the benefits of online shopping, such as convenience, safety, and trustworthiness, but also the risks that can occur to the average consumer.

To summarize, this research equips both businesses and consumers with the knowledge to navigate the future of the rapidly popularizing online shopping in Germany. Businesses can leverage these insights to enhance their strategies and customer engagement, while consumers can make better-informed choices, contributing to a more robust and dynamic online shopping environment.

8. Limitations and further research directions

To enhance the accuracy and reliability of future surveys and interviews, several improvements can be made. First, ensuring a more diverse and representative sample will help avoid sampling bias and ensure that the results reflect a wider population. Utilizing random sampling techniques and stratified sampling can achieve this diversity. Additionally, crafting clear, unbiased questions will minimize response bias and ensure that participants interpret the questions consistently. Incorporating both quantitative and qualitative questions can provide a more comprehensive understanding of participant experiences and opinions. To avoid social desirability bias, guaranteeing participant anonymity can encourage more honest and candid responses, this was however not made clear initially before conducting the surveys or interviews. Besides, more questions could have been incorporated and asked to gain a deeper understanding of both the participant and the research topic. An increased sample size would also gain a higher validity of data and findings, and conducting this survey in other areas could help reduce the majority of the participants being from southern Germany. This can be connected to environmental bias, while it was attempted to conduct the experiment under similar weather conditions in different locations, seemingly small factors such as noise, number of present people around, or simply time of the day could affect the reliability of the results. They could be well thought-out answers or were done in a rush due to either time constraint or any other factors. To accurately reduce environmental bias, knowing how to apply environmental filters to the data set reduces the effects of sampling bias. Filtering by environmental variables, rather than purely geographic ones, could lead to more accurate models (Varela et al., 2014).

A mixed methods approach can bring valuable insights of a topic that can sometimes not be interpreted simply through a linear methodological approach. However, to gain valuable lessons and utilize both quantitative and qualitative research results to the fullest, a lot of resources and time has to be invested (Teddlie & Tashakkori, 2009). Additionally, due to the drawbacks, quality may suffer in the data that is collected from both the survey and the interview. Surveys can result in lower response rates and interviews can lack in either interviewer skills or interviewee's openness for responses (Bryman, 2006).

The qualitative research, or the interview involved gathering extensive and detailed data, such as a transcription. Generally, analyzing this data is both time-consuming and labor-intensive to identify themes and patterns between participants. This required a thorough understanding of the data and often required multiple readings. The findings of the qualitative research was unreliable due to time constraints and inability to have a bigger sample size.

Investigator reflexivity refers to the process of researchers reflecting on their own biases, perspectives, and potential influences on the research process. A lack of reflexivity can impact research objectivity. A researchers' own beliefs, values, and experiences can unconsciously influence how the data is interpreted and interactions are done with the audience. Without reflexivity, there is a risk that researchers may selectively perceive and emphasize data that align with their preconceived notions, leading to skewed findings. In this research it is undeniable that some level of influence had affected the data collection process, due to multiple unstructured interactions having taken place.

Demand characteristics refer to cues that participants pick up from the research setting or researcher that influence their behavior or responses (Banyard et al., 1996). Participants may respond in ways they think are expected or acceptable to the researcher, rather than providing genuine answers. This can skew data towards an ethically correct expectation rather than a true genuine response. Subtle cues from the interviewer, such as body language or tone of voice, can unintentionally signal desired responses, leading participants to alter their answers accordingly. In an unstructured interview a lot of the information could have been affected due to the researcher.

In unstructured or semi-structured qualitative methods, the open-ended nature of questions can lead to the collection of large amounts of irrelevant information. Participants have shared extensive details that are not directly relevant to the research questions, making it difficult to sift through and identify pertinent data.

Addressing these challenges requires careful planning and methodological rigor in qualitative and even quantitative research. Strategies such as thorough training for researchers, using skilled facilitators for focus groups, conducting the research in a closed environment with less outside factors, and employing different techniques to encourage balanced participation can help mitigate or reduce these issues and enhance the reliability and validity of the research findings.

9. References

1. Acheampong, T. Y. (2021). Impact of Covid-19 on e-Commerce in the European Union. ENTRENOVA-ENTerprise REsearch InNOVAtion, 7(1), 86-95.
2. Lone, S., Harboul, N., & Weltevreden, J. W. J. (2021). 2021 European e-commerce report.
3. Abdelrhim, M., & Elsayed, A. (2020). The Effect of COVID-19 Spread on the e-commerce market: The case of the 5 largest e-commerce companies in the world. Available at SSRN 3621166.
4. Beckers, J., Weekx, S., Beutels, P., & Verhetsel, A. (2021). COVID-19 and retail: The catalyst for e-commerce in Belgium?. Journal of Retailing and Consumer Services, 62, 102645.
5. Paraschiv, D. M., ȚIțan, E., Manea, D. I., Ionescu, C. D., Mihai, M., & Șerban, O. (2022). The change in e-commerce in the context of the Coronavirus pandemic. Management & Marketing, 17(2), 220-233.
6. Campisi, T., Russo, A., Tesoriere, G., Bouhouras, E., & Basbas, S. (2021, September). COVID-19's effects over e-commerce: A preliminary statistical assessment for some European countries. In International Conference on Computational Science and Its Applications (pp. 370-385). Cham: Springer International Publishing.
7. Dannenberg, P., Fuchs, M., Riedler, T., & Wiedemann, C. (2020). Digital transition by COVID‐19 pandemic? The German food online retail. Tijdschrift voor economische en sociale geografie, 111(3), 543-560.
8. Zimmermann, V., & Köhler-Geib, F. (2023). Impact of the COVID-19 Pandemic on Different Groups of SMEs in Germany and Their Recovery. Intereconomics, 58(6), 333-341.
9. Amazon.com, Inc. (2020). *Amazon Q4 2020 Earnings Release.* https://s2.q4cdn.com/299287126/files/doc_financials/2020/q4/Amazon-Q4-2020-Earnings-Release.pdf
10. Amazon.com, Inc. (2020). *Amazon Q4 2023 Earnings Release.* https://s2.q4cdn.com/299287126/files/doc_financials/2023/q4/AMZN-Q4-2023-Earnings-Release.pdf
11. Almeida, F. (2018). Strategies to perform a mixed methods study. European Journal of Education Studies.
12. Dipl.-Ing. Annemarie Biermas (2019). Stadt Bensheim Demographiebericht 2019 Bevölkerungsprognose 2018 bis 2043 für die Gesamtstadt. (pp. 42-50) https://www.bensheim.de/fileadmin/media/bensheim/04-Leben-in-Bensheim/Demographie/Demographiebericht_2019_Bevoelkerungsprognose_2018-2043.pdf

13. Mullins, P. (1991). Tourism urbanization. International Journal of Urban & Regional Research, 15(3).
14. Murdoch, M., Simon, A. B., Polusny, M. A., Bangerter, A. K., Grill, J. P., Noorbaloochi, S., & Partin, M. R. (2014). Impact of different privacy conditions and incentives on survey response rate, participant representativeness, and disclosure of sensitive information: a randomized controlled trial. BMC medical research methodology, 14, 1-11.
15. Grote, M. H., Lo, V., & Harrschar–Ehrnborg, S. (2002). A value chain approach to financial centres–the case of Frankfurt. Tijdschrift voor economische en sociale geografie, 93(4), 412-423.
16. Plano Clark, V. L. (2017). Mixed methods research. The Journal of Positive Psychology, 12(3), 305-306.
17. Ivanov, D., & Das, A. (2020). Coronavirus (COVID-19/SARS-CoV-2) and supply chain resilience: A research note. International Journal of Integrated Supply Management, 13(1), 90-102.
18. Germany - ecommerce. International Trade Administration | Trade.gov. (2023, December 6). https://www.trade.gov/country-commercial-guides/germany-ecommerce
19. Brusch, M., & Stüber, E. (2014). Developments and classifications of online shopping behavior in Germany. *International Journal of Cyber Society and Education*, *7*(2), 147-156.
20. Ullrich, C. (2019). New Approach meets new economy: Enforcing EU product safety in e-commerce. *Maastricht journal of European and comparative law*, *26*(4), 558-584.
21. Sun, T. (2011). The roles of trust and experience in consumer confidence in conducting e‐commerce: a cross‐cultural comparison between France and Germany. *International Journal of Consumer Studies*, *35*(3), 330-337.
22. Piroth, Philipp & Rueger-Muck, Edith & Bruwer, Johan. (2020). Digitalisation in grocery retailing in Germany: an exploratory study. The International Review of Retail, Distribution and Consumer Research. 30. 1-19. 10.1080/09593969.2020.1738260.
23. Svetlana Fedoseeva & Ellen Van Droogenbroeck, 2023. "Pandemic pricing: Evidence from German grocery e‐commerce," Agribusiness, John Wiley & Sons, Ltd., vol. 39(4), pages 1139-1156, October.
24. Bartik, A. W., Bertrand, M., Cullen, Z., Glaeser, E. L., Luca, M., & Stanton, C. (2020). The impact of COVID-19 on small business outcomes and expectations. *Proceedings of the national academy of sciences*, *117*(30), 17656-17666.
25. Braun, S., & Osman, D. (2024). Online grocery shopping adoption versus non-adoption among the over-50s in Germany. *Electronic Commerce Research*, 1-38.

26. Fosso Wamba, S., & Akter, S. (2015). Big data analytics for supply chain management: A literature review and research agenda. In *Enterprise and Organizational Modeling and Simulation: 11th International Workshop, EOMAS 2015, Held at CAiSE 2015, Stockholm, Sweden, June 8-9, 2015, Selected Papers 11* (pp. 61-72). Springer International Publishing.
27. Mastari, M. (2020). APPLYING GENERIC STRATEGY IN CREATING CUSTOMER LOYALTY IN THE MIDDLE OF PANDEMIC VIRUS COVID-19 IN POLYCLINIC MCU XYZ CARDIOVASCULAR HOSPITAL. . https://doi.org/10.31933/dijms.v1i6.391.
28. Varela, S., Anderson, R., García‐Valdés, R., & Fernández‐González, F. (2014). Environmental filters reduce the effects of sampling bias and improve predictions of ecological niche models. Ecography, 37, 1084-1091. https://doi.org/10.1111/J.1600-0587.2013.00441.X.
29. Hardaker, S. (2022). More Than Infrastructure Providers–Digital Platforms' Role and Power in Retail Digitalisation in Germany. Tijdschrift voor economische en sociale geografie, 113(3), 310-328.
30. Heyvaert, M., Hannes, K., Maes, B., & Onghena, P. (2013). Critical appraisal of mixed methods studies. *Journal of mixed methods research*, *7*(4), 302-327.
31. Kunkel, S., Terhorst, S., & Beier, G. (2023). Digitalisation and resilience of industry sectors: a descriptive analysis of the COVID-19 crisis in Germany. International Journal of Technological Learning, Innovation and Development, 15(1), 48-72.
32. HDE Handelsverband Deutschland. (n.d.). E-Commerce-Umsätze. https://einzelhandel.de/presse/zahlenfaktengrafiken/861-online-handel/1889-e-commerce-umsaetze
33. Banyard, P., Grayson, A., & Orne, M. T. (1996). Demand characteristics. *Introducing psychological research: Sixty studies that shape psychology*, 395-401.
34. Bryman, A. (2006). Integrating quantitative and qualitative research: how is it done?. *Qualitative research*, *6*(1), 97-113.
35. Teddlie, C., & Tashakkori, A. (2009). *Foundations of mixed methods research: Integrating quantitative and qualitative approaches in the social and behavioral sciences*. Sage.
36. Davenport, T. H., Harris, J. G., Jones, G. L., Lemon, K. N., Norton, D., & McCallister, M. B. (2007). The dark side of customer analytics. *Harvard business review*, *85*(5), 37.

10. Appendix

Raw Data

Did you primarily live in Germany during the Covid-19 pandemic?	What is your gender?	What is your age group?	How frequently did you shop online before the COVID-19 pandemic?	Since the beginning of the COVID-19 pandemic, how has your overall frequency of online shopping changed?	Which product categories have you shopped for online BEFORE the pandemic?	Which product categories have you shopped for online DURING the pandemic?	What are the main reasons for any increase in your online shopping activity during the pandemic?	How has your average monthly spending on online shopping changed from before to during the COVID-19 pandemic?	Do you prefer shopping from local German online stores or international online stores?	After the pandemic, do you intend to continue shopping online?
Yes	Male	25-34	Monthly	Increased significantly	Groceries, Clothing, Food	Electronics, Groceries, Clothing, Pharmaceuticals, Home and Garden, Entertainment, Food	Convenience, Safety concerns	Increased significantly	No preference	Yes, the same as now
Yes	Male	18-24	Monthly	Increased significantly	Electronics, Food	Electronics, Groceries, Clothing, Pharmaceuticals, Entertainment, Food	Convenience	Increased slightly	International	Yes, less than now
Yes	Male	35-44	Weekly	Increased slightly	Clothing, Food	Electronics, Groceries, Clothing, Home and Garden, Entertainment, Food	Convenience, Safety concerns	Increased significantly	No preference	Yes, less than now
No										
No										

Yes	Female	35-44	Rarely	Increased significantly	Clothing, Entertainment	Clothing, Entertainment	Convenience, Safety concerns, Price Advantages	Increased significantly	No preference	Yes, the same as now
Yes	Male	18-24	Never	Increased slightly	Food	Electronics, Groceries, Clothing, Entertainment, Food	Convenience, Price Advantages	Increased significantly	No preference	Yes, less than now
No										
No										
Yes	Female	45-54	Rarely	Increased significantly	Clothing	Clothing, Pharmaceuticals, Entertainment, Food	Convenience, Safety concerns, More free time, Limited access to physical stores, Price Advantages	Increased significantly	Local	Yes, the same as now
No										
No										
Yes	Prefer not to say	65+	Monthly	Decreased significantly	Home and Garden, Entertainment	Home and Garden, Entertainment	Convenience, Safety concerns, More free time, Limited access to physical stores, Price Advantages	Increased significantly	International	Yes, more than now
Yes	Female	55-64	Monthly	Remained the same	Clothing	Electronics, Groceries, Clothing, Pharmaceuticals, Home and Garden, Entertainment, Food	Convenience, Safety concerns, More free time, Limited access to physical stores,	Increased slightly	International	Yes, the same as now

							Price Advantages			
Yes	Female	Under 18	Daily	Increased significantly	Entertainment, stationeries	Entertainment	Safety concerns	Has not changed	Local	Yes, the same as now
Yes	Female	65+	Never	Remained the same	None	Home and Garden	Convenience	Increased slightly	No preference	No, will shop in-person
Yes	Male	Under 18	Rarely	Increased slightly	Electronics, Clothing	Electronics, Clothing, Entertainment, Food	Convenience, Limited access to physical stores	Increased slightly	No preference	Yes, the same as now
Yes	Male	45-54	Rarely	Increased significantly	Clothing, Entertainment	Clothing, Entertainment	Convenience, Limited access to physical stores, Price Advantages	Increased significantly	No preference	Yes, the same as now
No										
No										
Yes	Female	35-44	Rarely	Increased significantly	Groceries, Clothing, Home and Garden, Food	Electronics, Groceries, Clothing, Home and Garden, Entertainment, Food	Convenience, Safety concerns, Limited access to physical stores, Price Advantages	Increased significantly	International	Yes, the same as now
Yes	Other	25-34	Monthly	Remained the same	Groceries, Entertainment, Food	Groceries, Clothing, Entertainment, Food	Convenience, Safety concerns, More free time, Limited access to physical stores, Price Advantages	Has not changed	International	Yes, the same as now

Yes	Female	18-24	Rarely	Increased slightly	Clothing, Pharmaceuticals	Groceries, Clothing, Pharmaceuticals, Food	Safety concerns, Limited access to physical stores	Increased slightly	No preference	Yes, less than now
Yes	Male	18-24	Rarely	Increased slightly	Food	Electronics, Clothing, Entertainment, Food	Convenience, Limited access to physical stores	Increased significantly	No preference	Yes, less than now
Yes	Other	18-24	Weekly	Increased slightly	Clothing, Food	Electronics, Groceries, Clothing, Entertainment, Food	Convenience, Safety concerns	Increased slightly	International	Yes, the same as now
Yes	Female	45-54	Rarely	Increased significantly	Food	Groceries, Clothing, Home and Garden, Food	Convenience, Safety concerns, Limited access to physical stores	Increased significantly	No preference	Yes, the same as now
Yes	Male	55-64	Never	Increased slightly	None	Electronics, Entertainment	Limited access to physical stores	Increased slightly	No preference	No, will shop in-person
Yes	Female	25-34	Rarely	Increased significantly	Home and Garden, Food, Baby Supplies	Groceries, Clothing, Home and Garden, Food, Baby Supplies	Safety concerns, Limited access to physical stores, Price Advantages	Increased slightly	Local	Yes, the same as now
Yes	Female	18-24	Never	Increased slightly	None	Groceries, Clothing	Safety concerns, Limited access to physical stores	Increased slightly	Local	Yes, less than now
Yes	Female	18-24	Monthly	Increased slightly	Clothing	Clothing, Pharmaceuticals, Food	Limited access to physical stores	Increased slightly	International	Yes, the same as now

Yes	Male	35-44	Rarely	Increased slightly	Electronics, Entertainment	Electronics, Entertainment, Food	Convenience, Limited access to physical stores	Increased slightly	Local	No, will shop in-person
No										
Yes	Female	18-24	Monthly	Increased significantly	Food	Electronics, Groceries, Clothing, Entertainment, Food	Convenience, Safety concerns, More free time, Limited access to physical stores, Price Advantages	Increased significantly	International	Yes, the same as now
Yes	Male	25-34	Rarely	Increased slightly	Food	Electronics, Groceries, Clothing, Entertainment, Food	Convenience, Safety concerns, Limited access to physical stores, Price Advantages	Increased slightly	Local	Yes, less than now
Yes	Male	25-34	Rarely	Increased significantly	Electronics, Food	Electronics, Groceries, Clothing, Entertainment, Food	More free time, Limited access to physical stores, Price Advantages	Increased significantly	International	Yes, the same as now
Yes	Prefer not to say	18-24	Never	Increased slightly	None	Groceries, Clothing	Safety concerns, Limited access to physical stores	Increased slightly	Local	No, will shop in-person
Yes	Prefer not to say	25-34	Monthly	Increased slightly	Clothing, Pharmaceuticals, Food	Groceries, Clothing, Pharmaceuticals, Food	Convenience	Increased slightly	No preference	Yes, the same as now

Yes	Female	65+	Never	Remained the same	None	None	Never shopped online before	Has not changed	No preference	No, will shop in-person
No										
Yes	Male	18-24	Monthly	Increased slightly	Pharmaceuticals, Entertainment, Food	Electronics, Groceries, Clothing, Pharmaceuticals, Home and Garden, Entertainment, Food	Convenience, Safety concerns, Limited access to physical stores	Increased slightly	International	Yes, the same as now
No										
No										
Yes	Female	55-64	Rarely	Increased significantly	Entertainment	Groceries, Clothing, Entertainment, Food	Convenience, Safety concerns, Limited access to physical stores	Increased significantly	International	Yes, less than now
Yes	Female	35-44	Monthly	Increased slightly	Clothing	Groceries, Clothing, Pharmaceuticals	Price Advantages	Increased significantly	No preference	Yes, the same as now
Yes	Male	45-54	Never	Increased slightly	None	Electronics, Groceries, Home and Garden	Price Advantages	Increased slightly	No preference	Yes, less than now
Yes	Female	35-44	Monthly	Increased slightly	Clothing	Clothing, Pharmaceuticals, Home and Garden	Convenience, Safety concerns	Increased slightly	Local	Yes, the same as now
No										
Yes	Male	Under 18	Never	Increased slightly	None	Clothing, Food	Convenience, Price Advantages	Increased slightly	International	Yes, the same as now
Yes	Male	25-34	Never	Increased slightly	None	Groceries, Food	More free time, Limited access to physical stores	Increased slightly	No preference	Yes, less than now

Yes	Male	25-34	Rarely	Increased significantly	Food	Electronics, Clothing, Entertainment, Food	Convenience, More free time	Increased significantly	No preference	Yes, the same as now
Yes	Female	45-54	Never	Increased significantly	None	Groceries, Clothing, Pharmaceuticals, Home and Garden	Safety concerns, Price Advantages	Increased significantly	Local	Yes, less than now
Yes	Male	18-24	Never	Increased slightly	None	Clothing, Food	Convenience	Increased slightly	No preference	Yes, the same as now
Yes	Female	18-24	Rarely	Increased slightly	Clothing	Clothing, Pharmaceuticals, Food	Convenience, Safety concerns	Increased slightly	No preference	Yes, the same as now
Yes	Female	25-34	Never	Increased significantly	None	Groceries, Clothing, Pharmaceuticals, Food	Safety concerns, Limited access to physical stores	Increased significantly	No preference	Yes, the same as now
Yes	Male	25-34	Monthly	Increased slightly	Food	Electronics, Clothing, Food	Convenience, Limited access to physical stores	Increased slightly	No preference	Yes, the same as now
Yes	Female	18-24	Rarely	Increased slightly	Clothing	Clothing, Pharmaceuticals, Food	Safety concerns, Price Advantages	Increased slightly	No preference	Yes, more than now
Yes	Female	18-24	Never	Increased slightly	None	Clothing, Pharmaceuticals	Safety concerns, Limited access to physical stores	Increased slightly	Local	Yes, the same as now
Yes	Male	18-24	Never	Increased significantly	None	Electronics, Clothing, Entertainment, Food	Convenience, Limited access to physical stores	Increased significantly	No preference	Yes, the same as now
Yes	Male	18-24	Rarely	Increased slightly	Electronics	Electronics, Clothing, Food	Limited access to	Increased slightly	No preference	Yes, the same as now

							physical stores			
Yes	Female	18-24	Rarely	Increased slightly	Clothing	Clothing, Pharmaceuticals, Food	Safety concerns	Increased slightly	No preference	Yes, the same as now
Yes	Female	18-24	Never	Increased significantly	None	Clothing, Pharmaceuticals, Home and Garden, Food	Safety concerns, Limited access to physical stores	Increased significantly	No preference	Yes, the same as now
Yes	Male	18-24	Rarely	Increased slightly	Food	Electronics, Clothing, Entertainment	Price Advantages	Increased slightly	No preference	Yes, the same as now
Yes	Female	18-24	Never	Increased slightly	None	Groceries, Home and Garden, Food	Safety concerns, Limited access to physical stores	Increased slightly	No preference	Yes, the same as now
Yes	Male	35-44	Rarely	Increased slightly	Electronics	Electronics, Pharmaceuticals, Home and Garden	Convenience, Price Advantages	Increased slightly	No preference	Yes, more than now
Yes	Female	18-24	Rarely	Increased significantly	Pharmaceuticals	Clothing, Pharmaceuticals, Home and Garden, Entertainment	Safety concerns, Limited access to physical stores	Increased significantly	No preference	Yes, the same as now
Yes	Male	25-34	Never	Increased significantly	None	Groceries, Clothing, Pharmaceuticals, Food	Safety concerns, Limited access to physical stores	Increased significantly	No preference	Yes, less than now
No										
Yes	Female	45-54	Rarely	Increased slightly	Entertainment	Clothing, Entertainment, Food	Safety concerns	Increased slightly	International	Yes, less than now
Yes	Male	18-24	Weekly	Increased slightly	Groceries, Clothing, Entertainment, Food	Electronics, Groceries, Clothing,	Convenience, Safety concerns, More free	Increased slightly	International	Yes, the same as now

						Entertainment, Food	time, Limited access to physical stores, Price Advantages			
No										
No										
No										
Yes	Male	25-34	Weekly	Increased significantly	Food	Electronics, Groceries, Clothing, Pharmaceuticals, Entertainment, Food	Convenience, Price Advantages	Increased slightly	No preference	Yes, the same as now
Yes	Male	25-34	Monthly	Increased significantly	Entertainment	Entertainment, Food	Convenience	Increased significantly	No preference	Yes, less than now
No										
No										
Yes	Female	Under 18	Never	Increased slightly	None	Clothing, Food	Limited access to physical stores, Price Advantages	Increased slightly	International	Yes, the same as now
Yes	Male	18-24	Monthly	Increased significantly	Clothing, Entertainment, Food	Electronics, Groceries, Clothing, Entertainment, Food	Convenience, Safety concerns, More free time, Limited access to physical stores, Price Advantages	Increased slightly	No preference	Yes, less than now
Yes	Male	35-44	Rarely	Increased significantly	Entertainment	Entertainment, Food	Convenience, Safety concerns	Increased slightly	No preference	Yes, the same as now

Yes	Prefer not to say	25-34	Weekly	Increased slightly	Groceries, Clothing, Entertainment, Food	Groceries, Clothing, Entertainment, Food	Convenience, Limited access to physical stores, Price Advantages	Increased slightly	International	Yes, the same as now
Yes	Female	35-44	Rarely	Remained the same	Groceries, Pharmaceuticals, Home and Garden	Groceries, Pharmaceuticals, Home and Garden	More free time, Price Advantages	Has not changed	No preference	Yes, the same as now
Yes	Male	25-34	Rarely	Remained the same	Food	Food	Price Advantages	Has not changed	No preference	Yes, the same as now
Yes	Female	18-24	Monthly	Increased slightly	Pharmaceuticals	Clothing, Pharmaceuticals, Food	Limited access to physical stores, Price Advantages	Increased slightly	International	Yes, the same as now
Yes	Male	18-24	Rarely	Increased slightly	Electronics	Electronics, Entertainment, Food	Convenience	Increased slightly	International	Yes, more than now
Yes	Female	18-24	Rarely	Remained the same	Clothing	Clothing	More free time	Has not changed	International	Yes, the same as now
No										
Yes	Male	25-34	Monthly	Decreased slightly	Electronics, Clothing, Entertainment, Food	Electronics, Clothing, Entertainment, Food	Price Advantages	Has not changed	International	Yes, the same as now
Yes	Male	65+	Never	Remained the same	None	None	None	Has not changed	No preference	No, will shop in-person
No										
Yes	Male	18-24	Weekly	Increased slightly	Entertainment, Food	Clothing, Entertainment, Food	More free time	Increased slightly	No preference	Yes, the same as now
Yes	Female	25-34	Monthly	Increased slightly	Clothing, Food	Electronics, Groceries, Clothing, Pharmaceuticals,	More free time, Limited access to physical	Increased significantly	International	Yes, less than now

						Entertainment, Food	stores, Price Advantages			
Yes	Male	18-24	Weekly	Decreased slightly	Electronics, Groceries, Clothing, Pharmaceuticals, Home and Garden, Entertainment, Food	Groceries, Pharmaceuticals, Entertainment	Reduced	Decreased slightly	International	Yes, the same as now
Yes	Female	35-44	Rarely	Remained the same	Clothing, Pharmaceuticals, Food	Groceries, Clothing, Pharmaceuticals, Food	Convenience	Increased slightly	No preference	Yes, the same as now
No										
Yes	Male	25-34	Never	Increased significantly	None	Electronics, Groceries, Clothing, Home and Garden, Entertainment, Food, None	Convenience, Safety concerns, More free time, Limited access to physical stores, Price Advantages	Increased significantly	No preference	Yes, less than now
No										
No										
Yes	Male	35-44	Monthly	Increased slightly	Electronics, Clothing, Pharmaceuticals	Electronics, Clothing, Pharmaceuticals, Entertainment, Food	Convenience, Limited access to physical stores, Price Advantages	Increased slightly	Local	Yes, less than now
Yes	Female	18-24	Rarely	Decreased slightly	Clothing	Clothing	Limited access to physical stores	Decreased slightly	International	Yes, less than now
No										

Yes	Male	35-44	Weekly	Increased slightly	Electronics, Pharmaceuticals	Electronics, Pharmaceuticals, Entertainment, Food	Convenience, Safety concerns, More free time, Price Advantages	Increased slightly	No preference	Yes, the same as now
Yes	Other	35-44	Monthly	Increased slightly	Clothing, Pharmaceuticals	Groceries, Clothing, Pharmaceuticals, Entertainment, Food	More free time	Increased slightly	International	Yes, the same as now
Yes	Female	55-64	Never	Increased slightly	None	Entertainment, Food	Convenience, More free time, Boredom	Increased slightly	No preference	Yes, the same as now
Yes	Female	25-34	Monthly	Decreased significantly	Electronics, Clothing, Home and Garden, Entertainment, Food	Entertainment	No increase	Decreased Significantly	No preference	No, will shop in-person
No										
No										
Yes	Male	55-64	Never	Remained the same	None	None	No increase	Has not changed	No preference	No, will shop in-person
No										
Yes	Female	18-24	Monthly	Increased slightly	Electronics, Groceries, Clothing, Pharmaceuticals, Home and Garden, Entertainment, Food	Electronics, Groceries, Clothing, Pharmaceuticals, Home and Garden, Entertainment, Food	Safety concerns, More free time, Limited access to physical stores	Increased slightly	Local	Yes, the same as now
Yes	Female	Under 18	Rarely	Increased slightly	Clothing, Entertainment	Clothing, Entertainment, Food	Convenience	Increased slightly	No preference	Yes, the same as now
Yes	Female	25-34	Rarely	Remained the same	Pharmaceuticals,	Pharmaceuticals,	Convenience, Price Advantages	Has not changed	International	Yes, the same as now

					Entertainment, Food	Entertainment, Food				
Yes	Male	25-34	Rarely	Increased slightly	Electronics, Food	Electronics, Clothing, Entertainment, Food	Limited access to physical stores, Price Advantages	Increased slightly	No preference	Yes, the same as now
Yes	Female	25-34	Rarely	Increased significantly	Food	Clothing, Pharmaceuticals, Home and Garden, Food	Limited access to physical stores	Increased significantly	International	Yes, the same as now
Yes	Male	18-24	Never	Increased slightly	None	Electronics, Groceries	Safety concerns	Increased slightly	No preference	Yes, more than now
Yes	Female	18-24	Monthly	Remained the same	Electronics, Pharmaceuticals	Electronics, Pharmaceuticals	Convenience, Price Advantages	Has not changed	International	Yes, the same as now
Yes	Female	18-24	Rarely	Increased significantly	Clothing	Clothing, Pharmaceuticals, Home and Garden, Entertainment	Safety concerns	Increased significantly	International	Yes, the same as now
Yes	Male	35-44	Rarely	Increased slightly	Electronics, Entertainment	Electronics, Groceries, Entertainment, Food	Safety concerns, Price Advantages	Increased slightly	No preference	Yes, the same as now
Yes	Female	35-44	Rarely	Increased slightly	Clothing	Clothing, Pharmaceuticals, Home and Garden, Entertainment	Safety concerns	Increased slightly	International	Yes, the same as now
Yes	Male	25-34	Rarely	Increased slightly	Entertainment	Electronics, Clothing, Entertainment	Convenience, More free time	Increased slightly	International	Yes, more than now
Yes	Female	35-44	Rarely	Increased significantly	Pharmaceuticals	Groceries, Pharmaceuticals, Home and Garden, Entertainment	Safety concerns	Increased significantly	International	Yes, the same as now
Yes	Female	25-34	Monthly	Remained the same	Clothing, Pharmaceuticals,	Clothing, Pharmaceutic	More free time	Has not changed	International	Yes, more than now

					Home and Garden	als, Home and Garden				
No										
Yes	Female	35-44	Rarely	Decreased slightly	Clothing, Entertainment	Entertainment	No increase	Decreased slightly	Local	Yes, the same as now
Yes	Male	65+	Never	Increased slightly	None	Groceries, Pharmaceuticals	Convenience, Safety concerns, Limited access to physical stores	Increased slightly	No preference	Yes, the same as now
No										
No										
Yes	Other	18-24	Monthly	Decreased slightly	Electronics, Clothing, Pharmaceuticals, Entertainment, Food	Clothing, Pharmaceuticals, Entertainment, Food	Convenience, Limited access to physical stores	Has not changed	No preference	Yes, less than now
Yes	Female	35-44	Never	Increased slightly	None	Food	Convenience, Safety concerns	Increased slightly	Local	Yes, more than now
Yes	Female	18-24	Rarely	Increased slightly	Food	Clothing, Pharmaceuticals, Food	Safety concerns, Limited access to physical stores	Increased slightly	International	Yes, more than now
Yes	Male	35-44	Never	Increased slightly	None	Electronics, Clothing, Entertainment	Limited access to physical stores	Increased slightly	International	Yes, the same as now
Yes	Male	18-24	Rarely	Increased slightly	Clothing, Food	Electronics, Clothing, Entertainment, Food	Convenience, Limited access to physical stores	Increased slightly	International	Yes, the same as now
Yes	Female	18-24	Rarely	Remained the same	Clothing, Pharmaceuticals	Clothing, Pharmaceuticals	Price Advantages	Has not changed	International	Yes, the same as now

Yes	Male	35-44	Rarely	Increased slightly	Entertainment	Electronics, Clothing, Entertainment	Limited access to physical stores, Price Advantages	Increased slightly	International	Yes, the same as now
Yes	Female	35-44	Monthly	Increased slightly	Clothing, Pharmaceuticals, Home and Garden	Groceries, Clothing, Pharmaceuticals, Home and Garden	Safety concerns, Limited access to physical stores	Increased slightly	No preference	Yes, less than now
Yes	Male	18-24	Rarely	Increased significantly	Entertainment	Clothing, Pharmaceuticals, Entertainment, Food	Safety concerns, Price Advantages	Increased significantly	No preference	Yes, less than now
Yes	Female	18-24	Never	Increased slightly	None	Clothing, Pharmaceuticals, Home and Garden	Safety concerns, Limited access to physical stores	Increased slightly	International	Yes, more than now
No										
No										
Yes	Female	18-24	Monthly	Increased slightly	Clothing	Clothing, Pharmaceuticals, Food	Safety concerns, Limited access to physical stores	Increased slightly	International	Yes, the same as now
Yes	Male	18-24	Rarely	Remained the same	Electronics, Clothing	Electronics, Clothing	Price Advantages	Has not changed	International	Yes, the same as now
Yes	Female	25-34	Monthly	Increased slightly	Entertainment, Food	Groceries, Home and Garden, Entertainment, Food	Safety concerns, Limited access to physical stores	Increased slightly	No preference	Yes, the same as now
Yes	Male	25-34	Rarely	Increased slightly	Pharmaceuticals, Entertainment	Pharmaceuticals, Home and Garden, Entertainment	Price Advantages	Increased slightly	International	Yes, the same as now

Yes	Female	35-44	Monthly	Remained the same	Groceries, Pharmaceuticals, Home and Garden	Groceries, Pharmaceuticals, Home and Garden	Price Advantages	Has not changed	Local	Yes, the same as now
Yes	Male	25-34	Rarely	Increased slightly	Electronics, Entertainment	Electronics, Clothing, Pharmaceuticals, Entertainment	Limited access to physical stores	Increased slightly	International	Yes, the same as now
Yes	Female	25-34	Rarely	Increased significantly	Clothing	Groceries, Clothing, Pharmaceuticals, Home and Garden	Safety concerns	Increased significantly	International	Yes, the same as now
Yes	Female	18-24	Rarely	Increased significantly	Clothing	Clothing, Pharmaceuticals, Home and Garden, Food	Safety concerns, Limited access to physical stores	Increased significantly	No preference	Yes, less than now
Yes	Male	45-54	Never	Increased slightly	None	Electronics, Entertainment, Food	Limited access to physical stores, Price Advantages	Increased slightly	No preference	Yes, the same as now
Yes	Female	35-44	Monthly	Remained the same	Groceries, Pharmaceuticals, Home and Garden	Groceries, Pharmaceuticals, Home and Garden	Price Advantages	Has not changed	No preference	Yes, the same as now
Yes	Male	35-44	Rarely	Increased slightly	Electronics	Electronics, Home and Garden, Food	Safety concerns, Price Advantages	Increased slightly	International	Yes, the same as now
No										
Yes	Female	35-44	Never	Increased slightly	None	Groceries, Pharmaceuticals	Safety concerns	Increased slightly	No preference	Yes, the same as now
No										
Yes	Male	25-34	Rarely	Increased slightly	Electronics, Clothing	Electronics, Clothing, Pharmaceutic	Limited access to physical	Increased slightly	International	Yes, less than now

						als, Home and Garden	stores, Price Advantages			
No										
Yes	Female	25-34	Never	Increased significantly	None	Groceries, Clothing, Pharmaceuticals, Home and Garden	Safety concerns, Limited access to physical stores	Increased significantly	Local	Yes, less than now
Yes	Male	25-34	Rarely	Increased slightly	Clothing	Electronics, Clothing, Entertainment	Price Advantages	Increased slightly	International	Yes, the same as now
Yes	Female	25-34	Rarely	Increased slightly	Pharmaceuticals	Groceries, Pharmaceuticals, Home and Garden	Safety concerns, Limited access to physical stores	Increased slightly	Local	Yes, less than now
Yes	Male	25-34	Monthly	Increased slightly	Entertainment, Food	Clothing, Entertainment, Food	Price Advantages	Increased slightly	No preference	Yes, more than now
Yes	Female	35-44	Never	Increased significantly	None	Clothing, Pharmaceuticals, Home and Garden	Safety concerns	Increased significantly	International	Yes, the same as now
Yes	Male	35-44	Never	Increased slightly	None	Groceries, Food	Limited access to physical stores	Increased slightly	Local	Yes, less than now
Yes	Female	25-34	Rarely	Increased slightly	Clothing	Clothing, Pharmaceuticals, Home and Garden	Safety concerns, Limited access to physical stores	Increased slightly	International	Yes, the same as now
No										
Yes	Male	25-34	Rarely	Increased slightly	Entertainment	Electronics, Entertainment, Food	Price Advantages	Increased slightly	No preference	Yes, the same as now
No										

Yes	Female	25-34	Monthly	Remained the same	Home and Garden, Food	Home and Garden, Food	Convenience	Has not changed	No preference	Yes, the same as now
Yes	Male	18-24	Rarely	Increased slightly	Clothing	Electronics, Clothing, Pharmaceuticals	Limited access to physical stores	Increased slightly	International	Yes, more than now
No										
Yes	Female	35-44	Monthly	Increased significantly	Groceries	Groceries, Pharmaceuticals, Home and Garden, Baby Supplies	Safety concerns, Limited access to physical stores	Increased significantly	No preference	Yes, less than now
Yes	Male	25-34	Rarely	Increased slightly	Electronics	Electronics, Pharmaceuticals, Home and Garden	Limited access to physical stores	Increased slightly	International	Yes, more than now
Yes	Male	35-44	Never	Increased slightly	None	Groceries	Safety concerns	Increased slightly	No preference	Yes, less than now
Yes	Female	25-34	Rarely	Increased significantly	Home and Garden, Entertainment	Groceries, Clothing, Pharmaceuticals, Home and Garden, Entertainment	Safety concerns, Limited access to physical stores	Increased significantly	No preference	Yes, less than now
Yes	Male	25-34	Rarely	Increased slightly	Entertainment	Electronics, Pharmaceuticals, Entertainment	Price Advantages	Increased slightly	International	Yes, the same as now
Yes	Female	18-24	Rarely	Increased slightly	Clothing	Clothing, Pharmaceuticals, Home and Garden	Limited access to physical stores	Increased slightly	International	Yes, the same as now
No										
Yes	Male	18-24	Never	Increased slightly	None	Groceries, Food	Limited access to physical stores	Increased slightly	No preference	Yes, less than now
Yes	Female	25-34	Rarely	Increased slightly	Clothing	Clothing, Pharmaceuticals, Home and Garden	Safety concerns, Limited access to	Increased slightly	International	Yes, the same as now

							physical stores			
Yes	Male	25-34	Rarely	Increased slightly	Food	Electronics, Clothing, Food	Limited access to physical stores, Price Advantages	Increased slightly	No preference	Yes, the same as now
Yes	Female	35-44	Monthly	Remained the same	Groceries, Pharmaceuticals	Groceries, Pharmaceuticals	Price Advantages	Has not changed	Local	Yes, the same as now
Yes	Male	35-44	Rarely	Increased slightly	Entertainment	Electronics, Home and Garden, Entertainment	Convenience, More free time	Increased slightly	International	Yes, more than now
No										
Yes	Female	25-34	Never	Increased significantly	None	Electronics, Clothing, Pharmaceuticals	Limited access to physical stores	Increased significantly	International	Yes, the same as now
Yes	Male	45-54	Never	Increased slightly	None	Groceries	Safety concerns	Increased slightly	No preference	No, will shop in-person
Yes	Male	18-24	Rarely	Increased slightly	Entertainment	Clothing, Entertainment, Food	Limited access to physical stores	Increased slightly	No preference	Yes, the same as now
No										
No										
Yes	Male	Under 18	Never	Increased slightly	None	Clothing	Limited access to physical stores, More options	Increased slightly	International	Yes, more than now
No										
Yes	Female	55-64	Rarely	Increased significantly	Entertainment	Electronics, Groceries, Clothing, Entertainment	Safety concerns, More free time, Limited access to	Increased significantly	Local	Yes, the same as now

							physical stores			
Yes	Female	55-64	Monthly	Remained the same	Clothing, Pharmaceuticals	Clothing, Pharmaceuticals	Convenience	Increased slightly	No preference	Yes, less than now
No										
No										